

| Question | Answer                                                                                         | Marks     |
|----------|------------------------------------------------------------------------------------------------|-----------|
| 2(a)     | $pV = nRT$<br>$T = (5.60 \times 10^5 \times 3.80 \times 10^{-2}) / (5.12 \times 8.31)$         | <b>C1</b> |
|          | $T = \mathbf{500\text{ K}}$                                                                    | <b>A1</b> |
| 2(b)(i)  | $V/T$ is constant<br>$V = (3.80 \times 10^4) \times (500 + 125) / 500$                         | <b>C1</b> |
|          | $V = \mathbf{4.75 \times 10^4\text{ cm}^3}$                                                    | <b>A1</b> |
| 2(b)(ii) | (for ideal gas,) change in internal energy is change in (total) kinetic energy (of molecules)  | <b>B1</b> |
|          | $\Delta U = 3/2 \times 1.38 \times 10^{-23} \times 125 \times 5.12 \times 6.02 \times 10^{23}$ | <b>C1</b> |
|          | $= \mathbf{7980\text{ J}}$                                                                     | <b>A1</b> |
| 2(c)(i)  | $w = p\Delta V$<br>$= 5.60 \times 10^5 \times (4.75 - 3.80) \times 10^{-2}$                    | <b>C1</b> |
|          | $= \mathbf{5320\text{ J}}$                                                                     | <b>A1</b> |
| 2(c)(ii) | total $= 7980 + 5320$<br>$= \mathbf{13300\text{ J}}$                                           | <b>A1</b> |

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|----------|------------------------------------------------------|-------|
| 2(c)     | roughly shaped circle or oval surrounding the origin | B1    |